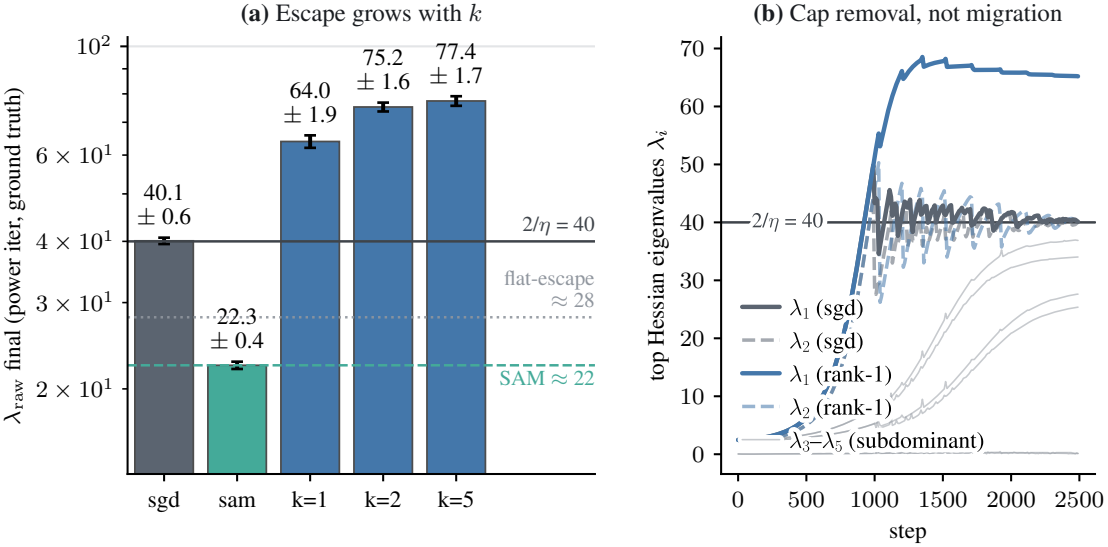

(a) Escape grows with k
λ_{raw} final (power iter, ground truth)
10^2
6×10^1
4×10^1
3×10^1
2×10^1
40.1 ± 0.6
22.3 ± 0.4
64.0 ± 1.9
75.2 ± 1.6
77.4 ± 1.7
$2/\eta = 40$
flat-escape ≈ 28
SAM ≈ 22
sgd
sam
k=1
k=2
k=5
(b) Cap removal, not migration
top Hessian eigenvalues λ_i
70
60
50
40
30
20
10
0
$2/\eta = 40$
λ_1 (sgd)
λ_2 (sgd)
λ_1 (rank-1)
λ_2 (rank-1)
λ_3=λ_5 (subdominant)
0
500
1000
1500
2000
2500
step